MHT correction cuts both ways: evidence from the Graduation programme

Banerjee et al. (2015) · Viviano, Wüthrich & Niehaus (2026)

The Graduation programme evaluation pools six countries and ten outcome families, offering a natural MHT problem when one outcome is tested across J=6 country-specific treatment arms. When costs per arm are mostly variable — as they are in a multi-site field experiment — the VWN framework implies a relatively permissive threshold. The key lesson is that correction can either remove marginally-significant results or restore ones that step-up FDR procedures discard.

One outcome × J=6 country arms

$$Y_i^k = \alpha + \sum_{c \in country}^{6} \beta_c \, assignment_i \times T_c + \beta Z_i^k + U_{\text{shortsurveys}} + V_{\text{stratification}} + \varepsilon_i$$

We replace the single treatment dummy with six country-specific interactions. Each of the ten outcome families is tested against J=6 hypotheses. VWN optimal thresholds ($\bar{\alpha}$=0.05, one-sided) are α=0.0210 (Cobb-Douglas) and α=0.0401 (Linear), both two-sided equivalents. The J=6 correction is considerably milder than J=10: each country arm cost millions in implementation, making heavy correction economically unjustified.

Significant country arms by selected outcomes — endline 1 (out of J=6)

Outcome	Naive 10%	BH/FDR	VWN-CD	VWN-Lin
Income	5	5	5	5
Consumption	4	4	3	4
Food security	4	4	2	4
Physical health ⚑	5	3	1	2
Mental health	2	2	2	2
Women's empowerment	1	1	1	1

⚑ *Physical health: 5 naive → 1 under VWN-CD. Only India's effect survives (t=2.38, p=0.017).*

VWN-Lin rejects where BH does not: physical health at follow-up

Physical health at endline 2 is the clearest illustration that correction can cut the other way. BH rejects zero countries; VWN-Lin rejects one.

Country	Coeff	SE	p (2s)	Naive 10%	BH/FDR	VWN-CD	VWN-Lin
Ethiopia	+0.0393	0.0506	0.438	✗	✗	✗	✗
Ghana	−0.0136	0.0547	0.804	✗	✗	✗	✗
Honduras	+0.0349	0.0399	0.382	✗	✗	✗	✗
India	−0.0047	0.0501	0.926	✗	✗	✗	✗
Pakistan	−0.0023	0.0471	0.961	✗	✗	✗	✗
Peru ⚑	+0.1007	0.0469	0.033	✓	✗	✗	✓

Peru's p-value (0.033 two-sided) passes VWN-Lin's threshold (α=0.040) but fails BH's step-up rule, which requires p(1) ≤ 0.05/6 = 0.0083 when only one out of six p-values is marginal. BH is conservative precisely when a single hypothesis is isolated near the threshold.

Calibrating the linear cost function to Table 4

Cost decomposition

Table 4 Panel A itemises per-participant costs in PPP dollars for each country arm. Nearly all observable line items scale with J: livestock/asset transfers, staff time, food support, travel, and training. The only plausibly J-fixed cost — study-wide coordination — is unobserved but small relative to per-arm implementation spending of $1,000–$5,000 per household.

Cost category	Share	Classification
Staff, assets, food, travel, materials	~66%	Variable with J
Training & start-up	~8%	Per-arm (scales with J)
Indirect + other supervision	~23%	Ambiguous — treated as fixed (upper bound)
Study-wide coordination	unobserved	J-fixed (small, unobserved)

Conservative calibration treats all ambiguous items as fixed, giving cf = 0.23. This yields α* = 0.023 one-sided (0.047 two-sided) — well above VWN defaults, because costs are overwhelmingly per-arm.

Where calibrated VWN and BH diverge

In 18 of 20 outcome-period cells, calibrated VWN and BH agree. The two exceptions:

- **L_cal rejects more than BH — Physical health EL2**: *Peru (p_1□=0.016) passes the calibrated threshold (0.023) but fails BH's step-up rule (requires p(1) ≤ 0.0083 when only one country is marginal). Robust from cf=0 through cf=0.40.*

- **BH rejects more than L_cal — Income EL2**: *BH retains all 6 countries via step-up cascade.* L_*cal misses Peru* ($p_{1\square}$=0.033) *and Pakistan* ($p_{1\square}$=0.049), *both above the* 0.023 *threshold.*

Key takeaways

1. **Correction is not one-directional.** VWN-CD strips most of Physical Health's country-level significance at EL1 (5→1). But VWN-Lin's calibrated threshold recovers Peru's Physical Health effect at EL2 that BH discards.
2. **The appropriate correction depends on cost structure.** FDA-style defaults (cf=0.46) are calibrated to pharma fixed-cost economics. In a multi-site field experiment where each arm independently costs millions, the economically justified threshold is far more permissive — close to an uncorrected 5% test.

Data: Banerjee et al. (2015) Science 348(6236). Method: Viviano, Wüthrich & Niehaus (2026) Stata package. Regression: Eq. 2 (country-specific treatment dummies), clustered SEs at randomisation unit. VWN thresholds at $\bar{\alpha}$*=0.05 one-sided.*